

Electron Transitions and the Hydrogen Spectrum

SimuPhysics Atomic Model & Hydrogen Spectrum Simulation · SimuPhysics

Years 9-10

~30 minutes

Science · Chemistry

NOTICE — AI-generated worksheet

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Simulation: <https://simuphysics.com/view/modern-physics-atomic-model-hydrogen-spectrum-simulation.html>

Curriculum links: AC9S10U06 · AC9S9U06 · atomic structure

Learning intentions

- I can describe how electrons move between energy levels in an atom.
- I can explain how electron transitions produce specific lines in an emission spectrum.
- I can connect atomic structure to the historical development of the atomic model.

Getting started

Open the SimuPhysics Atomic Model & Hydrogen Spectrum Simulation. Trigger an electron transition (moving an electron between energy levels) and watch for the corresponding line appearing in the spectrum display. Try several different transitions before starting the tasks.

Tasks

1. Trigger a transition where the electron drops from a higher energy level to a lower one. What happens on the spectrum display when this occurs?

2. Trigger a transition between two energy levels that are far apart, then one where the levels are close together. Which transition produces higher-energy (e.g. more towards blue/violet) light?

3. Trigger four different electron transitions and record the colour or position of the spectral line each one produces.

Transition (levels)	Colour/position of spectral line

4. Explain why the hydrogen spectrum shows only specific lines, not a continuous rainbow of colours. What does this tell you about electron energy levels inside the atom?

5. Based on your table, does a bigger jump between energy levels produce more or less energetic light? Explain your reasoning.

6. Describe how the discovery of the atomic emission spectrum contributed to scientists moving away from earlier models of the atom toward the idea of fixed electron energy levels.

7. If an electron absorbs energy and jumps to a higher level, then later drops back down, does it release the same amount of energy it absorbed? Explain using the simulation.

8. Explain how the position of an element in the periodic table relates to the number and arrangement of its electrons, using hydrogen as your simplest example.

Extension (optional)

Each element produces its own unique emission spectrum, like a fingerprint. Explain how astronomers could use this idea to work out what a distant star is made of, without ever travelling there.
